

Major project progress report

Patient-Independent Localization of Myocardial Infarction from Vectorcardiographic Signals Using Tensor Decomposition and Machine Learning

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Project question

Can a VCG-based myocardial infarction (MI) localization method that scores highly on random heartbeat splits also classify **completely unseen patients**? We reproduced the published tensor-decomposition baseline, changed the evaluation unit to patients, investigated stronger representations and models, and tested transportability to another dataset.

Results at a glance

Experiment	Cohort / protocol	Main result
Paper-style reproduction	PTB; 178 patients; random beat 10-fold CV	99.14% beat accuracy
Unseen-patient test	PTB; 178 patients; patient-grouped 10-fold CV	41.39% beat accuracy
Best six-class PTB model	PTB; 164 patients; nested patient CV	0.487 patient macro F1
External validation	PTB to PTB-XL; 1,092 held-out patients	0.242 patient macro F1

Main finding: the paper-style beat-level score is nearly reproduced, but it does not translate to comparable performance on unseen patients. The current system is a research prototype, not a clinically reliable diagnostic model.

What has been completed

An end-to-end, reproducible codebase now includes PTB data preparation; wavelet denoising and R-peak detection; VCG beat tensor construction and Tucker features; random-beat and patient-independent evaluation; nested six-class model benchmarks; 12 representation ablations; a strict PTB-XL cohort audit; external testing; and a paired Frank-versus-Kors VCG domain-shift experiment.

What remains

The planned improvement phase - deep signal models and tensor/deep hybrid models - has **not** been implemented. No state-of-the-art or clinical-performance claim is currently justified. The supervisor should review the six-class endpoint, label mapping, and next experiment before model development.

1. Paper reproduction and the generalization gap

The reference study by Zhang et al. (IEEE TBME, 2023) combines wavelet-based VCG tensors, Tucker compression and a TreeBagger classifier. It reports **99.80% accuracy** on normal and 11 MI-location categories in the PTB database [1]. Our implementation follows that pipeline using Frank X/Y/Z signals.

Processing stage	Implemented choice
Signal preparation	1,000 Hz Frank XYZ; bior6.8 denoising; Pan-Tompkins-style R-peak detection
Beat and tensor	651 samples per beat (250 before, 400 after R); 3 leads x 651 time x 8 signal/subband slices
Tensor feature	D3-D9 wavelet details; Tucker time rank 3; 72 features per beat
Classifier	500-tree bootstrap random forest as the TreeBagger analogue

Cohort audit

The matched PTB cohort has **178 subjects** (126 MI, 52 healthy), 425 VCG recordings and 60,507 detected beats. The paper reports 60,527 beats - a difference of 20 (0.033%). Exact beat annotations and several preprocessing details were unpublished, so this is a close reimplementa-tion rather than a bitwise replication.

Evaluation changes the conclusion

Evaluation	Patient overlap	Accuracy	Macro F1
Random beat 10-fold CV	Same patients can appear in train and test	99.14%	0.991
Patient-grouped 10-fold CV	No patient shared across train and test	41.39%	0.191
Published paper result	Beat-level protocol	99.80%	Not directly comparable

The accuracy difference between our two protocols is 57.75 percentage points. This is evidence of severe within-patient dependence in the beat-level task, not proof that the original authors intentionally introduced leakage.

Why we changed the primary task

Several of the 12 classes have only one independent patient. Holding out that patient leaves no training example of the class. We therefore defined a six-class endpoint with at least ten patients per class: AMI, ALMI, ASMI, IMI, ILMI and healthy controls (HC). It contains 164 patients and 55,863 beats; patient-level macro F1 is the primary metric.

2. Patient-independent model and feature studies

The six-class benchmark uses fixed patient-disjoint outer folds, inner patient-disjoint tuning, patient-balanced training weights and patient-level aggregation of beat probabilities. A healthy-only model reaches 31.71% patient accuracy on PTB; accuracy alone is not sufficient.

Model / representation	Patients	Accuracy	Macro F1	Macro ROC-AUC
Tucker + random forest	164	47.56%	0.380	0.764
Tucker + linear SVM	164	43.90%	0.366	0.767
Tucker + XGBoost	164	46.95%	0.403	0.770
Wavelet statistics + XGBoost	164	54.88%	0.487	0.817

Representation ablation

We evaluated 12 representations on the same frozen PTB patient folds: Tucker ranks 1/2/3/5/8; raw versus denoised beats; amplitude normalization; individual leads; no wavelet subbands; and wavelet statistics without Tucker compression. The latter was strongest in the exploratory comparison. Its 192 features summarize each lead and subband with eight statistical descriptors of shape, amplitude and energy.

Nested confirmation

With the same nested selection protocol, wavelet statistics increased patient macro F1 from **0.403** to **0.487**, and patient accuracy from **46.95%** to **54.88%**. The paired bootstrap interval for the macro-F1 difference is +0.003 to +0.165 (10,000 class-stratified patient resamples).

Important qualification: the representation was selected after viewing exploratory results on these PTB folds. The nested rerun helps compare model selection but is not an independent confirmation of the representation choice.

What this means for the project title

Tensor decomposition remains the reproduced baseline and a central scientific comparator. The strongest current six-class PTB result comes from wavelet statistics **without** Tucker compression. This is an informative finding, not a reason to relabel the paper baseline as the improved model.

3. External validation and domain-shift analysis

PTB-XL v1.0.3 provides 12-lead ECG rather than measured Frank VCG. We converted I, II and V1-V6 to estimated XYZ using the Kors regression, resampled from 500 to 1,000 Hz, and applied one fixed PTB-trained wavelet-statistics model to fold 10 without PTB-XL tuning [2, 3].

Strict PTB-XL test cohort	Value
Records / independent patients	1,185 / 1,092
Patients per class	AMI 19; ALMI 5; ASMI 104; IMI 154; ILMI 24; HC 786
Patient accuracy / macro F1	56.23% / 0.242
Balanced accuracy / macro ROC-AUC	24.21% / 0.685

The strict cohort accepts one target MI code or isolated NORM, excludes ambiguous/non-target MI records, and removes patients whose records have conflicting status. This improves label specificity but does not make PTB and PTB-XL diagnoses equivalent.

External result interpretation

The cohort is 72.0% HC. An always-HC prediction would obtain 71.98% accuracy, above our 56.23%; the low 0.242 macro F1 is therefore the more informative summary. ALMI has only five test patients and zero correct classifications. The model is **not** suitable for diagnostic use.

Paired PTB control: measured versus derived VCG

To isolate the VCG transformation from other differences, we compared simultaneous ECG and Frank recordings for the same 391 PTB records. We held patient folds, labels and 55,863 beat positions constant.

Training to testing domain	Patient accuracy	Patient macro F1
Measured to measured	52.44%	0.469
Measured to Kors-derived	47.56%	0.422
Kors-derived to Kors-derived	51.83%	0.467

Median record-level measured-versus-derived signal correlation was X 0.740, Y 0.821, and Z 0.584. The median matched-feature correlation was 0.869.

The measured-to-derived macro-F1 change is -0.046, with paired bootstrap 95% interval -0.108 to +0.017. It is consistent with a transform mismatch, but the interval includes zero. The much larger PTB-to-PTB-XL loss cannot be assigned to this factor alone; device, 10-second duration, label definitions and case mix also change.

4. Current contribution, limitations and next phase

Current research contribution

The project demonstrates that a near-reproduced high beat-level result is not evidence of unseen-patient reliability; provides a patient-disjoint six-class benchmark; tests a concrete feature improvement; and documents failure to transport well to a second dataset. All of these are completed experimental results, including the negative findings.

Limitations to state in the viva/report

Issue	Implication
Small PTB class cohorts	The primary six-class task has 164 patients; uncertainty remains large.
Selection on PTB folds	Wavelet-statistics gain is promising but needs a new untouched confirmation set.
PTB versus PTB-XL mismatch	Measured versus derived VCG and clinical labels differ; scores are not controlled comparisons.
Sparse external classes	Only five ALMI patients in the PTB-XL fold-10 strict test cohort.
No clinical validation	The code is for research, not deployment or diagnosis.

Proposed next milestone - not yet completed

Freeze the target labels, patient splits and primary metric with the supervisor. Then implement a compact 1D CNN/ResNet VCG baseline and a tensor-plus-learned-feature hybrid, using patient-disjoint model selection and the same held-out-patient comparison. Only claim a substantial improvement if paired uncertainty and a genuinely fresh external evaluation support it. Published PTB-XL deep-learning numbers are for different label tasks and cannot be used as direct SOTA comparators [4].

Reproducibility and meeting handoff

Repository: github.com/Kumaryan12/major_project. The README contains setup and command sequences; all experiment configurations and machine-readable metrics are checked in. Raw datasets, cached features and fitted models are not committed. Latest verified state: commit a939096, clean working tree before this report, and 15 passing automated tests.

Suggested decisions for the guide meeting: confirm the six-class patient-level endpoint; review the strict PTB-XL label mapping; agree on the deep-model comparison and available compute; decide what independent validation will support the final claim.

References

- [1] Zhang et al., IEEE Transactions on Biomedical Engineering 70(3), 812-823 (2023), DOI: 10.1109/TBME.2022.3202962. PubMed record.
- [2] Wagner et al., PTB-XL v1.0.3, PhysioNet. Dataset and fold protocol. [3] Tereshchenko Laboratory, Kors transform implementation.
- [4] Strodthoff et al., Deep Learning for ECG Analysis: Benchmarks and Insights from PTB-XL. Reference benchmark repository.